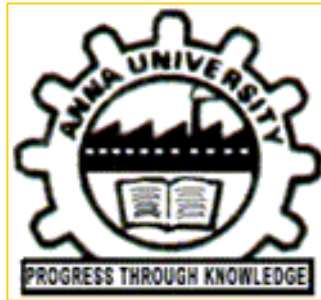


MADRAS INSTITUTE OF TECHNOLOGY

ANNA UNIVERSITY

CHROMPET

CHENNAI – 600 044



EC23CO2 ANALOG CIRCUITS DESIGN

{LAB INTEGRATED THEORY COURSE}

July 2024 to November 2024

PROJECT REPORT ON

SCHMITT TRIGGER

SUBMITTED BY

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In partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

BONAFIDE CERTIFICATE

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ANALOG CIRCUITS DESIGN a theory cum lab course during period **JULY 2024 –**
NOVEMBER 2024.

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Signature of the Professor

ABSTRACT

This project focuses on designing a Schmitt Trigger circuit using a BC547 transistor to convert a sinusoidal input signal into a square wave and also to process a square wave output into a sinusoidal waveform. A Schmitt Trigger is an electronic circuit that provides hysteresis, meaning it has two distinct threshold levels—one for switching the output high and one for switching it low. This property makes the circuit ideal for cleaning up noisy or fluctuating signals.

The circuit converts a sinusoidal input into a square wave by switching the output between high and low states when the input signal crosses the defined threshold voltages. This ensures that any noise or small variations in the input signal do not affect the output, producing a clean and stable square wave.

Additionally, the output square wave can be processed through a low-pass filter to remove the higher frequency components, converting the square wave back into a smooth sinusoidal signal. The resistor network in the circuit sets the threshold levels for the hysteresis, helping to create stable transitions and avoid false triggering. This project demonstrates the Schmitt Trigger's usefulness in waveform shaping, signal conditioning, and noise filtering in electronic systems.

In this mini project, we aim to design and build a Square Wave to Sine Wave Converter using basic electronic components like resistors and capacitors. Square waves are commonly used in digital systems, but they contain unwanted high-frequency harmonics that can cause distortion, especially in audio and communication applications. The purpose of this project is to convert the harsh square wave into a smooth sine wave, which is more desirable for such applications.

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Introduction:

Schmitt Trigger

A **Schmitt Trigger** is a special type of comparator that helps clean up noisy analog signals by introducing **hysteresis**. It ensures a stable digital output (high or low) even if the input signal fluctuates or has noise. The hysteresis is achieved by setting two different voltage thresholds, one for switching the output high and another for switching it low. This prevents the output from switching rapidly due to small variations in the input.

Operation: The Schmitt Trigger compares the input signal to these threshold voltages. When the input exceeds the upper threshold, the output switches to a high state, and when it drops below the lower threshold, the output switches to a low state.

Square wave to sine wave converter

In many electronic applications, especially in audio processing, square waves can produce unwanted harmonics. The goal of this section of the project is to convert a square wave into a smooth sine wave, which is more suitable for audio signals.

Operation: The square wave input will be passed through a filtering circuit that will smooth out the sharp edges of the square wave, removing high-frequency harmonics and creating a sine wave-like output. Filters, such as low-pass filters, will be used to achieve this transformation.

Aim:

To design and build an Schmitt Trigger circuit that converts a noisy or slowly changing input signal into a sharp, stable output signal with a defined threshold. This project will help demonstrate how hysteresis works to clean up noisy signals and create a clear transition between different voltage levels.

Components Required:

S.NO	COMPONENTS	RANGE	QUANTITY
1.	Resistor	10k , 100 1.5k 1k	1 Each 3 4
2.	Capacitor	10nF	3
3.	Transistor	BC547	2
4.	Function Generator	-	1
5.	DSO	-	1
6.	Bread Board	-	1
7.	LED	-	2
8.	Wires	-	As Required

Hardware Description:

Transistor:

The BC547 is an NPN transistor used for low-power switching and amplification. Key specs: max voltage 45V, max current 100mA, and power dissipation 500mW. It has 3 pins: Collector, Base, and Emitter. Commonly used in signal amplification, switching, and oscillator circuits.

Resistors:

1. 10k Ω : Often used in voltage dividers, signal conditioning, and biasing circuits.
2. 1k Ω : Common in signal filtering, current limiting, and as part of voltage dividers.
3. 1.5k Ω : Similar to the 1k Ω resistor, often used in biasing, feedback loops, or as a part of signal conditioning circuits.
4. 100 Ω : This lower resistance value is commonly used for current limiting, in series with LEDs, or for power distribution to limit the flow of current.

Capacitor:

A 10nF capacitor is a capacitor with a capacitance value of 10 nanofarads (nF). In electronic circuits, capacitors store and release electrical energy. Capacitors with values like 10nF are commonly used for decoupling, filtering, and timing applications in circuits.

DSO:

A DSO (Digital Storage Oscilloscope) is an instrument used to capture, display, and analyze electronic signal waveforms in digital form. It offers features like waveform storage, digital processing, and measurement tools for accurate signal analysis.

Function Generator:

A Function Generator is an electronic device used to generate different types of waveforms, such as sine, square, triangle, and sawtooth, at various frequencies and amplitudes. It is commonly used in testing and designing circuits by providing input signals for analysis or simulation.

Working:

Schmitt Trigger:

A Schmitt Trigger is a type of comparator circuit used to convert noisy or slowly varying analog signals into clean, stable digital signals. The key feature of a Schmitt Trigger is hysteresis, which introduces two distinct threshold voltage levels for switching the output: one for the high state and another for the low state. When the input voltage rises above the upper threshold, the output switches to a high state (e.g., 5V). When the input drops below the lower threshold, the output switches to a low state (e.g., 0V). This hysteresis ensures that small fluctuations or noise in the input signal do not cause unwanted output changes. Essentially, it prevents rapid switching caused by noise between the high and low levels. Schmitt Triggers are widely used in digital systems for signal conditioning, noise filtering, and providing clean logic signals, especially when dealing with unstable or noisy input signals.

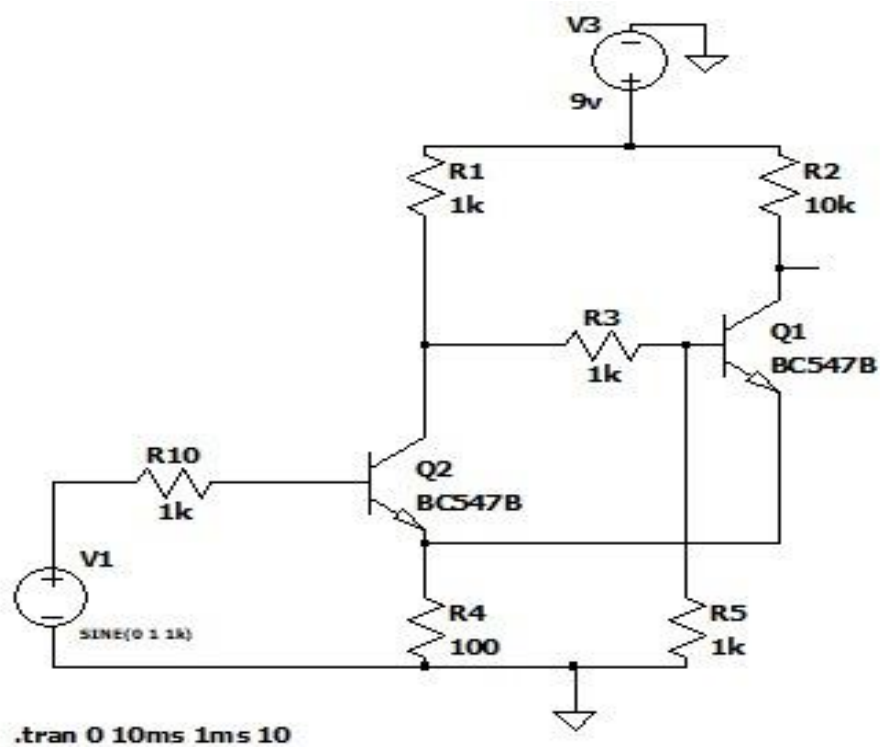
Square wave to sine wave converter:

A Square Wave to Sine Wave Converter uses resistors and capacitors to transform a square wave into a smooth sine wave. The square wave signal passes through the RC circuit, where the resistor limits current, and the capacitor filters out high-frequency components, including sharp transitions. This smoothing process removes the unwanted harmonics from the square wave, resulting in a cleaner sine wave output. By adjusting the resistor and capacitor values, the filter's cutoff frequency can be controlled to produce a more accurate sine wave for applications like audio signal processing.

Circuit Diagram and its Output:

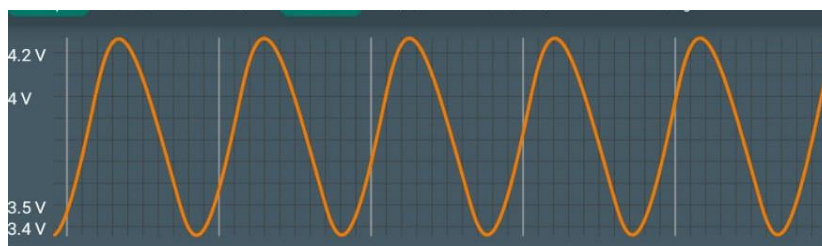
Schmitt Trigger:

Circuit diagram

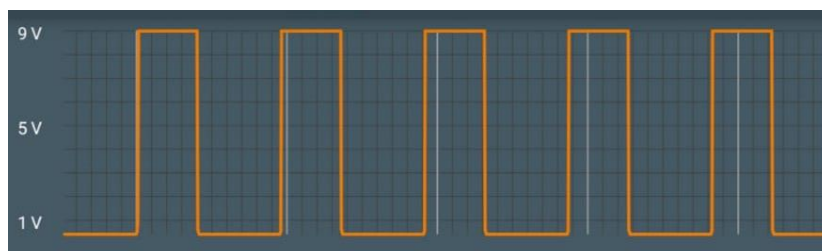


Output

Input Wave:

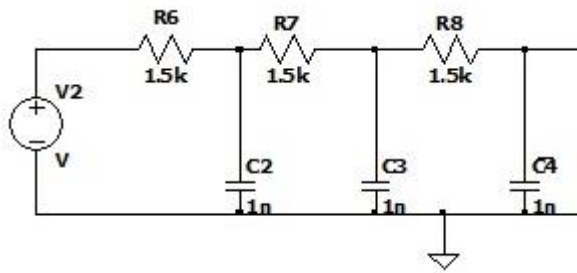


Output Wave:



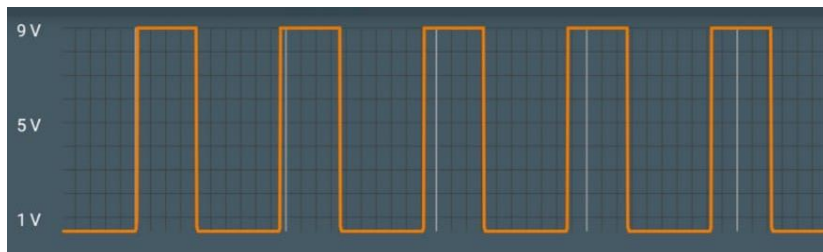
Square wave to sine wave converter:

Circuit diagram

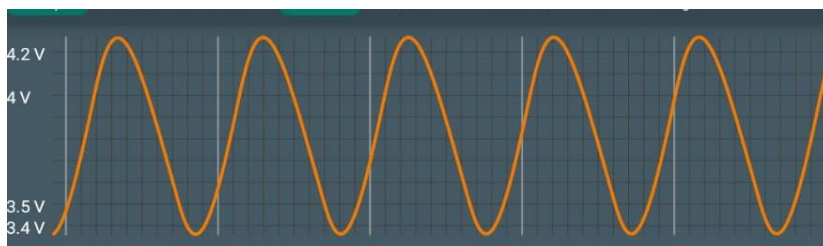


Output

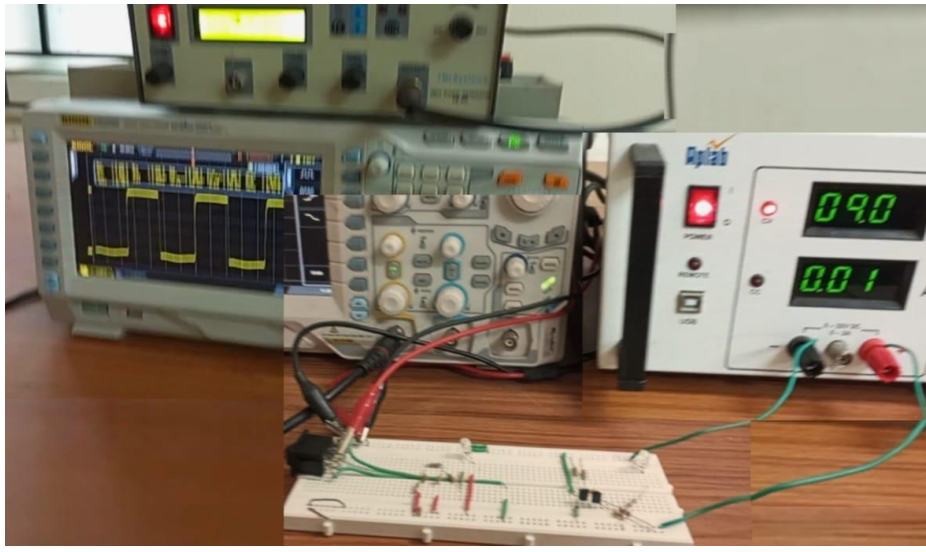
Input wave:



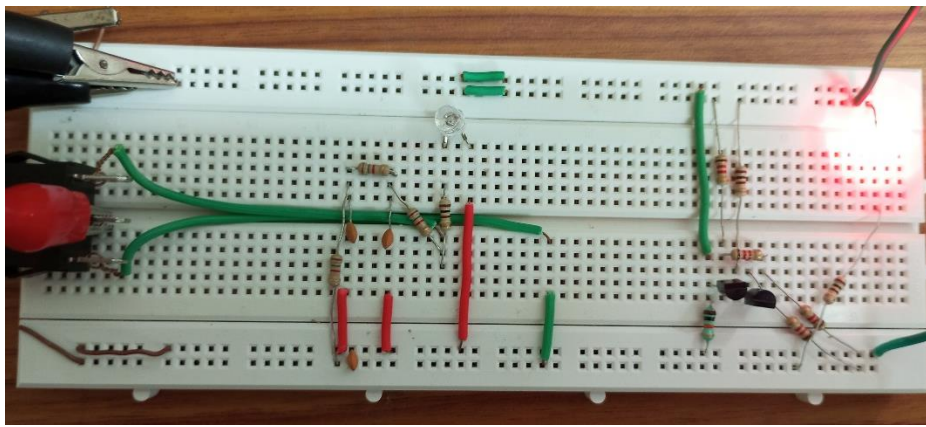
Output Wave:



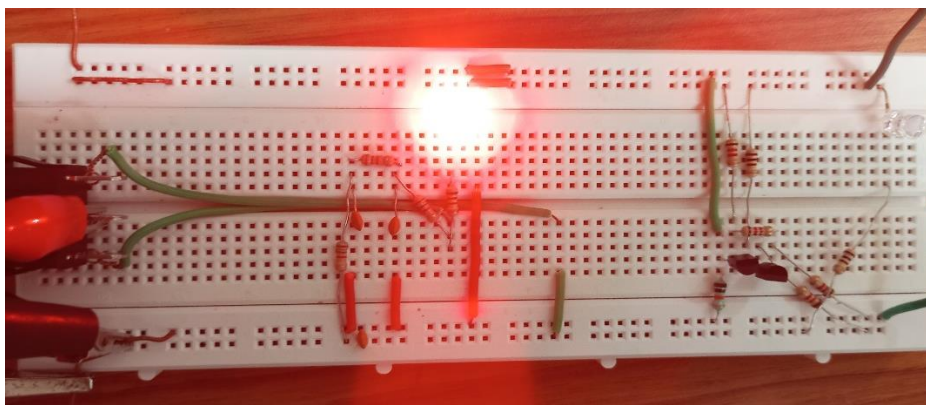
Experimental setup:



Activation of Schmitt Trigger:



Activation of Square Wave to Sine Wave Converter:



Applications:

SCHMITT TRIGGER:

1. **Noise Filtering:** Eliminates noise from analog signals, converting them into clean, stable digital outputs.
2. **Signal Conditioning:** Ensures stable digital signals from noisy or slowly changing analog inputs.
3. **Switch Debouncing:** Cleans up mechanical switch signals to avoid multiple detections of a single press.

SQUARE WAVE GENERATOR:

1. **Clock Generation:** Provides consistent timing signals for digital systems and microcontrollers.
2. **Pulse Width Modulation (PWM):** Used in motor control and power regulation by adjusting the width of square pulses.
3. **Signal Testing:** Serves as a test signal in electronics for analysing circuit behaviour and performance.

Advantages:

1. **Noise Immunity:** Rejects small fluctuations or noise, ensuring stable output.
2. **Stable Output:** Provides a clean, distinct square wave even with noisy input.
3. **Prevents False Triggering:** Two threshold levels prevent unwanted switching from noise or minor input variations.
4. **Simple Design:** Easy and inexpensive to design with basic components.

Conclusion:

In this project, we explored the design and functionality of two important electronic circuits: the Schmitt Trigger and the Square Wave to Sine Wave Converter. The Schmitt Trigger proved to be an effective tool for converting noisy or fluctuating analog signals into clean, stable digital outputs by utilizing hysteresis. This makes it invaluable in applications requiring noise filtering, signal conditioning, and stable digital transitions. On the other hand, the Square Wave to Sine Wave Converter, using a simple RC low-pass filter, demonstrated how to smooth sharp square wave transitions and remove high-frequency components, producing a clean sine wave ideal for audio and other applications requiring smooth waveforms.

Both circuits are essential in various practical applications, such as in signal processing, communication, and audio systems, ensuring better signal integrity and performance. This project highlights the significance of these fundamental components in modern electronics, showing how basic components like resistors, capacitors, and op-amps can be effectively used to manipulate and improve signal quality.

Reference:

1.Schmitt trigger is referred from ,

"The Art of Electronics" by Paul Horowitz and Winfield Hill
A classic textbook in electronics, it includes a detailed section on comparators and Schmitt Triggers, explaining their design, behaviour, and use cases in digital systems.

2. Square wave to sine wave converter is referred from,

<https://dcaclab.com/blog/how-to-build-square-wave-to-sine-wave/>